INCH-POUND
MIL-S-24187/9A(SH)
22 March 1991
SUPERSEDING
MIL-S-24187/9(SH)
9 September 1983

MILITARY SPECIFICATION SHEET

SWITCH ASSEMBLIES, LINEAR MOVEMENT, MANUAL AND REMOTE OPERATION, STYLE 5DLS

This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24187.

INCHES	MM
.001	0.03
.005	0.13
.03	0.76
.12	3.05
.28	7.11
.312	7.93
.45	11.43
3.060	77.72
4.00	101.63
7.060	179.32
7.96	202.18
9.40	238.76
9.96	252.98
14.00	355.60
16.45	417.83
16.67	423.42

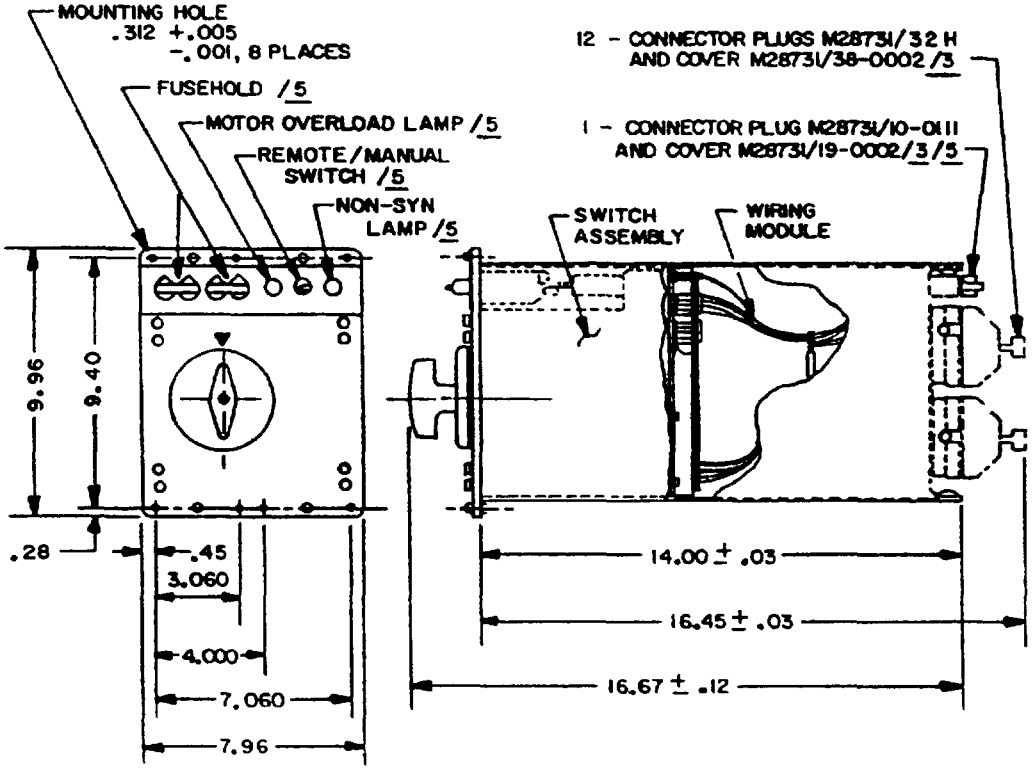


FIGURE 1 *Style 5DLS switch assembly*

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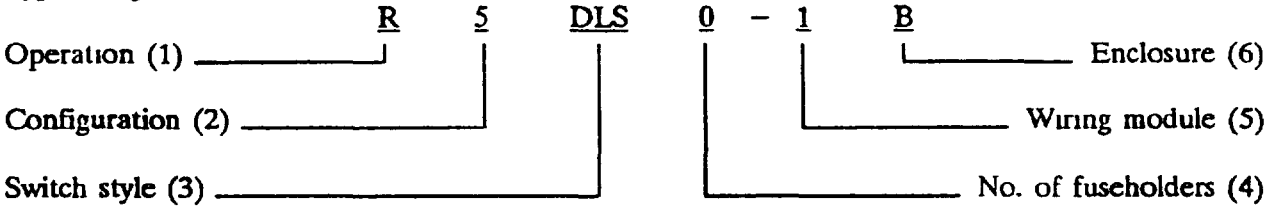
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NOTES:
1. Dimensions are in inches
2. Unless otherwise specified, tolerances are plus or minus 0.02 (0.51 mm) for two-place decimals and plus or minus 0.005 (0.13 mm) for three-place decimals.
3. Connector plugs with covers shall be supplied only when specified.
4. For suggested mounting see Drawing 815-5002994.
5. Supplied only on remote operated types.
6. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. *Style 5DLS switch assembly – Continued.*

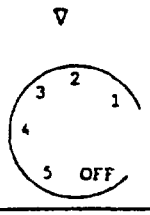
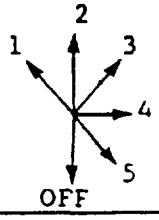
Type designation:



(1) Switch assemblies shall be remote operated with manual override, designated R, or manually operated, designated M.
(2) Contact configuration shall be type 5 (see figure 2).
(3) Switch style shall be designated by the letters DLS.
(4) There shall be no provisions for additional fuseholders.
(5) Wiring module basic circuitry shall be indicated by the designator 1 as specified to accomplish the switching actions specified in table I.
(6) Enclosed switch assemblies shall be indicated by a single letter (B).

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TABLE I. *Switch selection.*

Type designation						Application data						
Switch operation	Contact configuration	Switch style	Additional fuseholders	Wiring module	Enclosure	Switching action[1]	Stop position	Dial position[2]	Handle position[3]	Wired contacts[4]	To ship wiring connection[5]	Unwired contacts[4]
M or R	5	DLS	0	-1	B	ON1-5, OFF	6&8			( )E LOAD 5 ( )D LOAD 4 ( )C LOAD 3 ( )B LOAD 2 ( )A LOAD 1 ( )X INPUT	JK, JN JJ, JM JH, JL JD, JG JC, JF JB, JE	( )F OFF

Example: Type designation for the standard remote operated switch assembly shown with enclosure is R5DLS0-1B; the manual version is M5DLS0-1B

[1]Stops shall be adjustable to limit switch handle rotation to the positions indicated or to any consecutive positions within the maximum rotation indicated.

[2]Reference position of a style 5DLS switch is considered to be that in which the flat on the handle shaft is down. The mark ∇ and the dial positions shown are used to designate the reference position (CAUTION - this is different than other switches).

[3]Normal position of a style 5DLS switch is the position which coincides with normal circuit conditions in any application. Handle shall be adjustable to point upward. The handle and rotary description plate rotate.

[4]See figures 2 and 3 for complete description of contact identification and table V for load switch wiring.

[5]See figures 4 and 5 and tables V and VI for connector identifications and arrangements.

REQUIREMENTS:

Construction: Style 5DLS switch parts, modules and assemblies shall be constructed in accordance with Drawing 815-5002994.

Characteristics: Characteristics shown in table II shall apply to the style 5DLS switch assemblies.

TABLE II. *Characteristics.*

Characteristics	Value
Total number of poles available	150
Wired load switching poles	150
Number of positions	6
Break-before-make	Yes
Make-before-break	No
Electrical rating and endurance loading	
2 Amperes, 15 Vdc	20 000 Operations
25 Microamperes, 1.5 Vdc	20 000 Operations

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TABLE II. *Characteristics – Continued*

Characteristics	Value
Remote operation:	
Transfer time:	
Minimum	0.010 second
Maximum	0.070 second
Actuating time:	
1 Position change	0.15 – 0.40 second
5 Position change (one direction)	0.75 – 2.00 second
Current drain:	
at 115 V, 3ϕ, 60 Hz	1 Ampere rms per phase max. (0 70 Amp typ.)
Contact resistance:	
At 2 Amperes	0.025 ohm
At 2.5 Microamperes	0 05 ohm
Total circuit resistance:	
At 2 Amperes	0.080 ohm
At 25 Microamperes	0.115 ohm
Torque – shaft rotation maximum – Initial	90 pound-inches
After shock	100 pound-inches
– stop strength	150 pound-inches
Maximum weight:	
Enclosed with wiring module	40 pounds
Without enclosure and wiring module	30 pounds
Temperature and humidity (for rotational torque and endurance tests)	
Minimum	Plus 40 °F (plus 4 °C)
Maximum	Plus 149 °F (plus 65 °C) and 40 to 50 percent rh
Shock	MIL-S-901, grade A, class I, type C
Vibration	MIL-STD-167-1, type I, except omit variable frequency
Temperature rise	30 °C

Switch assembly modules: Style 5DLS manually operated switch assemblies shall consist of the following modules as detailed on Drawing 815-5002994.

Switch assembly
- Front module
- Mechanism module
- Switch contact module

Wiring module

Enclosure

Remote operated, style 5DLS switch assemblies shall have, in addition to the above, the

Remote control module/mechanism module (associated circuitry)

Control circuit module

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Switch function: Style 5DLS switch assemblies shall provide the number of poles indicated – each consisting of seven stationary contacts X, A, B, C, D, E, and F and a movable contact arranged as shown on figures 2 and 3. A stop (screw) shall prevent transfer of the movable contact into position X. Additional stops (removable) shall be provided which, when inserted, will limit the rotation to six consecutive positions or less, as required. Stops shall be detailed on Drawing 815-5002994 and shall be accessible from the front of the switch assembly, after removal of the handle and rotary description plate.

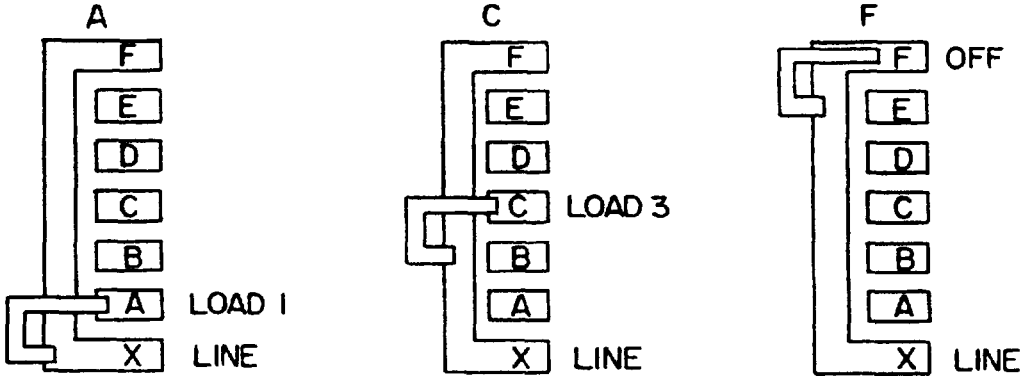


FIGURE 2. *Schematic typical operation of pole 1.*

TABLE III. *Stop positions versus switching action (full range of stops).*

Stop position	Switch position accessible	Switching action
6 and 8	A-B-C-D-E-F	1-2-3-4-5-OFF

Contact designation and marking: Each stationary switching contact shall be identified by a number, two letters and a second number. The first number shall identify the switch section (1 or 2) in which the contact is located. The first letter, A, B, C, D, or E shall identify the pole group in which a contact is located and the second letter shall be assigned as shown on figures 2 and 3. The second number shall be that of the column in which a contact is located (see figure 3). Each contact in the switch contact module connector plug shall be assigned the same identification as the contact in the switch contact module to which it connects. Contact F is common with contact X in all poles. Contacts shall be installed as shown on figure 3. Circles signify omitted contacts; phantom lines indicate mating header plates.

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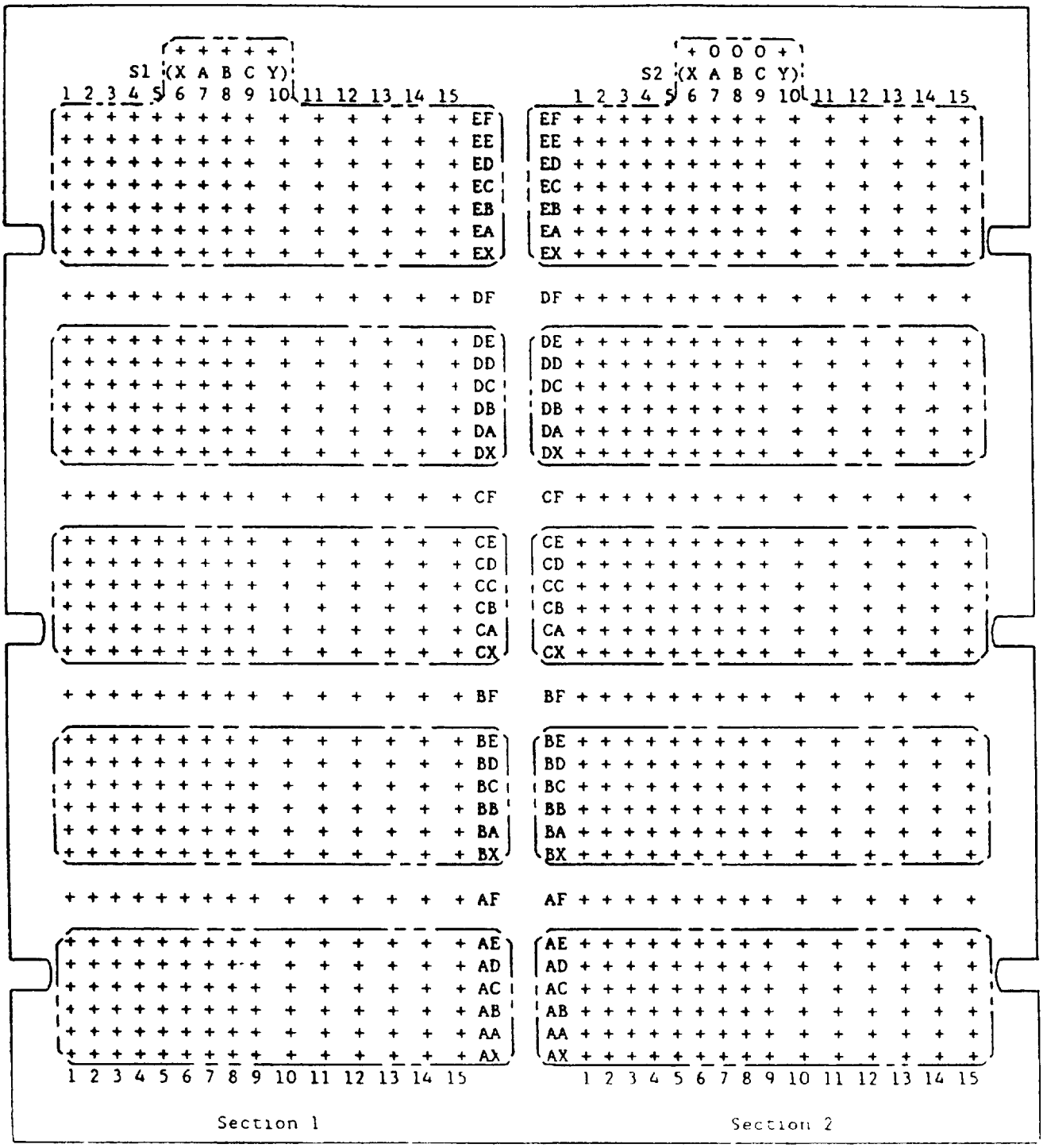
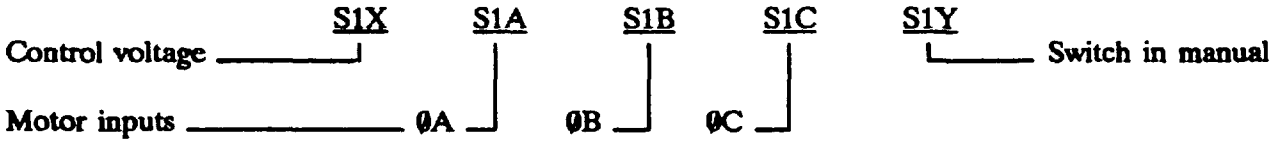


FIGURE 3. *Contact designation and marking, rear view*

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Remote operation: Connections to the switch module for remote operation shall be made to section 1 contacts as follows:



Remotely operated switch assembly shall move to the positions shown in table IV when the specified control voltage is applied.

TABLE IV. *Control voltage for remote operation.*

Control voltage	Closed switch contacts	XFMR tap[1,2]
0	X-A	A
16.4	X-B	B
32.8	X-C	C
49.2	X-D	D
65.6	X-E	E
82.0	X-F	F

[1]See figure 3 of MIL-S-24187 for typical test circuits for remote operated switch assemblies. Test loading is shown on figure 7 of this document.

[2]See figure 2 for typical pole positions corresponding to the transformer tap voltages A through F.

Enclosure and wiring module: The enclosure and the wiring module shall be furnished to connect the switch contact module to the external mating connectors at the rear of the enclosure (see figure 5). The general arrangement of the wiring shall be as shown on figure 4. The interconnecting wiring shall be as shown in tables V and VI.

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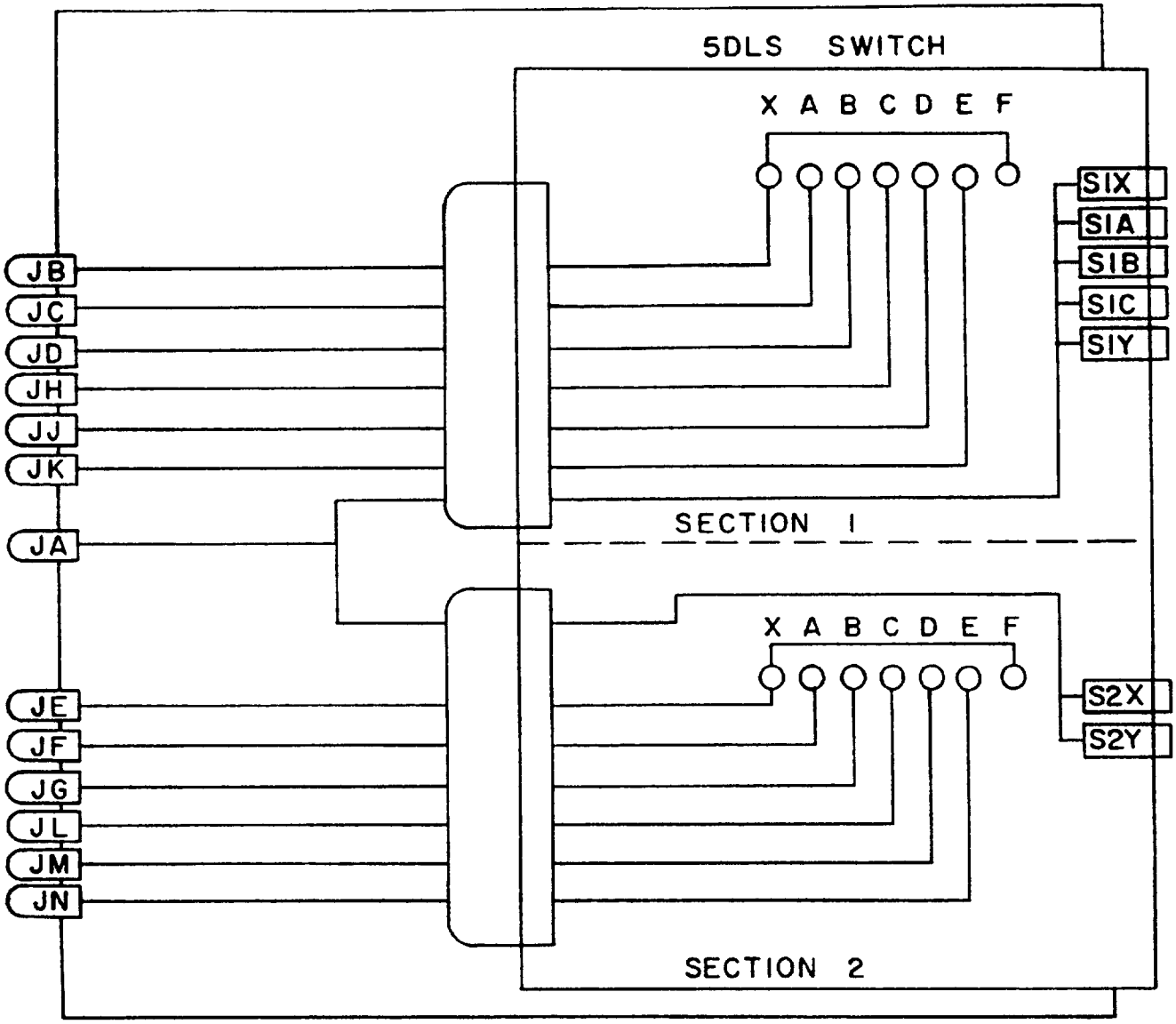


FIGURE 4. *Interconnecting wiring arrangement schematic.*

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Wiring module connections:

Unless otherwise specified in the contract or order, wiring modules shall provide wiring for 150 load switching poles as specified in table V. Wired contacts in each of the 150 poles shall be as speci-fied in table I. Module wiring for switch control and indicating circuits shall be in accordance with table VI and figure 6. Wire shall be type B-22, multi-strand, in accordance with MIL-W-16878. The color coding shall be as illustrated in the example below. Cable shields and grounds may be connected to any unused connector points.

TABLE V. *Wiring module connections, 117 pin connectors.*

_[1]	Switch contact module contact														
	Pole column														
Pole group	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
E	C2	B2	A2	A3	B3	C3	C4	B4	A4	A5	B5	C5	C6	B6	A6
D	F2	E2	D2	D3	E3	F3	F4	E4	D4	D5	E5	F5	F6	E6	D6
C	N3	N2	M2	K2	J2	H2	J3	H4	J5	H6	J6	K6	M6	N6	
B	R2	S2	T2	T3	S3	R3	R4	S4	T4	T5	S5	R5	R6	S6	T6
A	U2	V2	X2	X3	V3	U3	U4	V4	X4	X5	V5	U5	U6	V6	X6
	117 Pin connector terminals														

[1]Each pole group contains switch contacts X, A, B, C, D, E and F (see figures 2 and 4) Contact F is unwired and is common to contact X.

EXAMPLE: Pole E 5 that is connected to pin B3 is accordance with table V is illustrated below for all connections.

Section	Contact	Connector	Wire
S1	X	JB	Black
	A	JC	White
	B	JD	Red
	C	JH	Blue
	D	JJ	Orange
	E	JK	Yellow
	F	None	None
S2	X	JE	Black
	A	JF	White
	B	JG	Red
	C	JL	Blue
	D	JM	Orange
	E	JN	Yellow
	F	None	None

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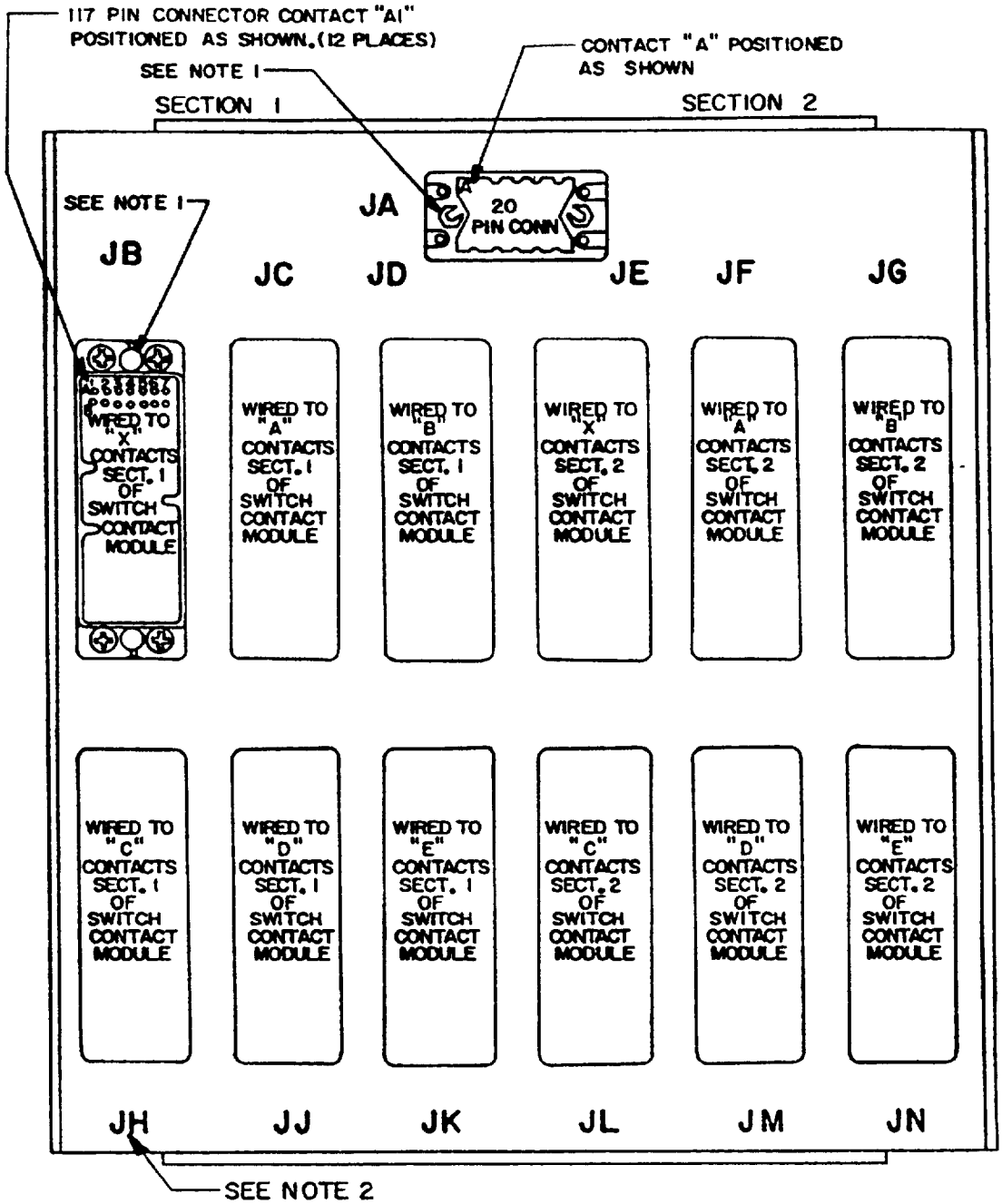
Wiring module continued:

The 20 pin connector shall be wired to the specified switch contacts as shown in table VI using gray wire. Switch module contacts S1A, S1B, S1C, S1X and S1Y are for the remote control circuit; S2X and S2Y are for the blown fuse indicator (BFI) circuit (see figure 6).

TABLE VI. *Wiring module connections, 20 pin connector.*

20 Pin connector JA contact	From JA to switch contact module	
A	S1A	
B	S1B	
C	S1C	
D	S1X	
E	S1Y	
V	S2X	
W	S2Y	
P	CX15 (S2)	Status circuits
N	CA15 (S2)	
M	CB15 (S2)	
U	CC15 (S2)	
T	CD15 (S2)	
S	CE15 (S2)	
J	CX15 (S1)	
H	CA15 (S1)	
F	CB15 (S1)	
L	CC15 (S1)	
R	CD15 (S1)	
K	CE15 (S1)	

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NOTES.

1 On connector JA polarizing guide sockets shall be in position 1 (as shown) On connectors JB through JN polarizing guide pins shall be in position 4 (as shown)

2 Characters are 0.19 inch high, black epoxy ink Locate to be visible with connector covers in place

FIGURE 5. *Enclosed switch connector arrangement, rear view*

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Remote control circuit: Control circuit for remote operation shall be in accordance with figure 6.

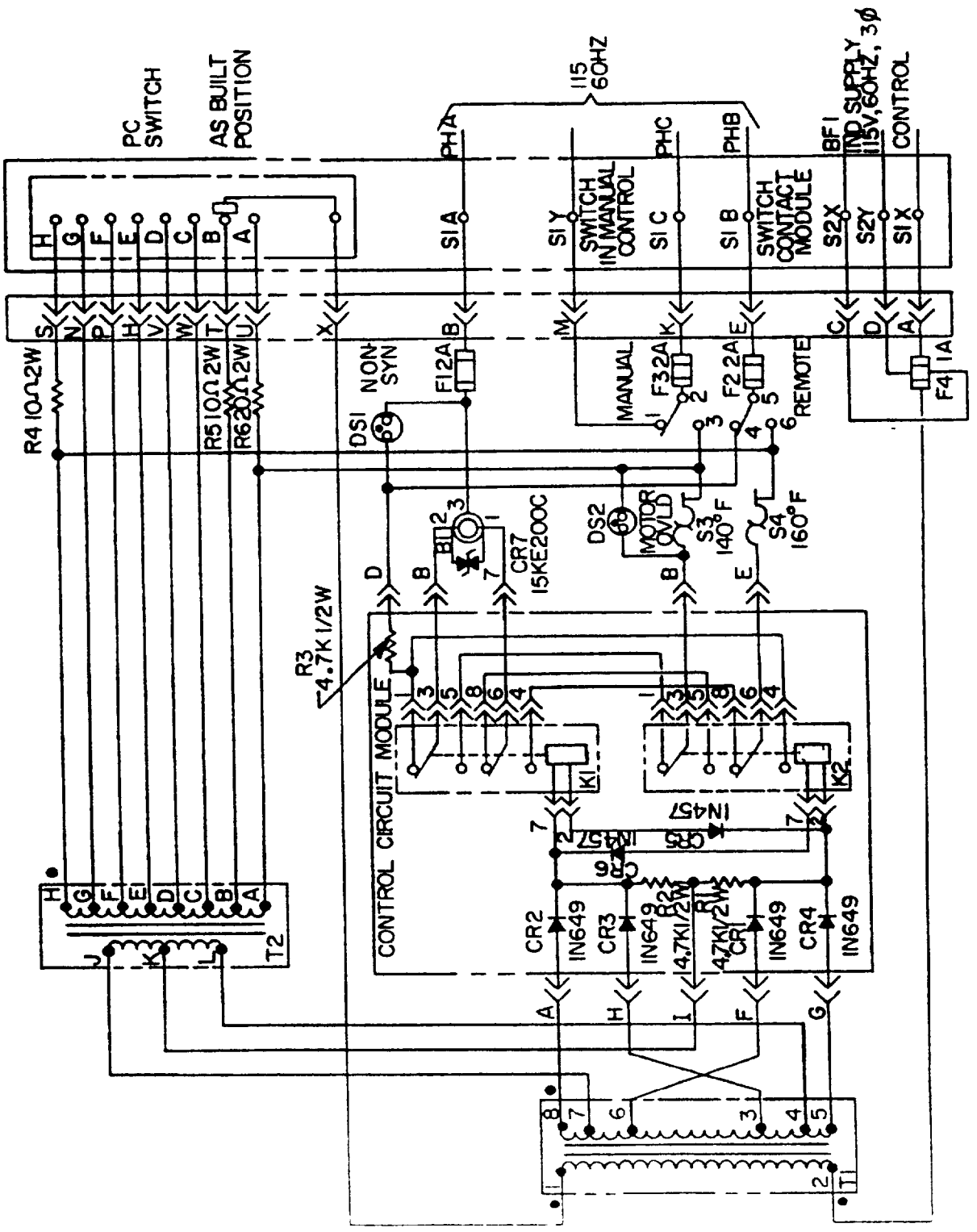


FIGURE 6. *Remote control circuit.*

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Repair parts: Repair parts shall be as specified in table VII

TABLE VII. *Repair parts*

Drawing 815-5002994 piece No.		Description	Quantities required for quantity of switches installed					
			1–3	4–12	13–30	31–60	61–120	121 or more
Manual	Remote							
	1	Enclosed switch,	1	1	1	1	2	2
	158	Stop screw	1	1	1	2	2	3
	32	Detent spring	1	1	1	2	3	6
	87	Toggle switch	1	1	1	2	3	4
	85	Indicator light	1	1	1	2	3	5
	86	Indicator light	1	1	1	2	3	5
		Lamp (M15098/11-001)	Note 2					
	10	Control circuit module	1	1	1	2	2	3
	93	Relay	1	2	3	Note 3		
	160	Washer, insul	2	2	2	4	6	8
	159	Spacer, insul.	2	2	2	4	6	8
	81	Fuseholder	1	1	1	2	3	4
		Fuse carrier (1A0981)	1	1	2	3	4	5
	82	Fuse, 2.0 ampere	Note 4					
	83	Fuse, 1.0 ampere	Note 4					

NOTES:
1. Refer to Drawing 803-4680149 for piece numbers.
2. One hundred percent repair lamps shall be provided
3. Supply one for each 10 or fraction of 10 installed
4. One hundred fifty percent repair fuses shall be supplied Fuses shall be supplied in multiples of five, in standard commercial packages, with a minimum quantity of 15

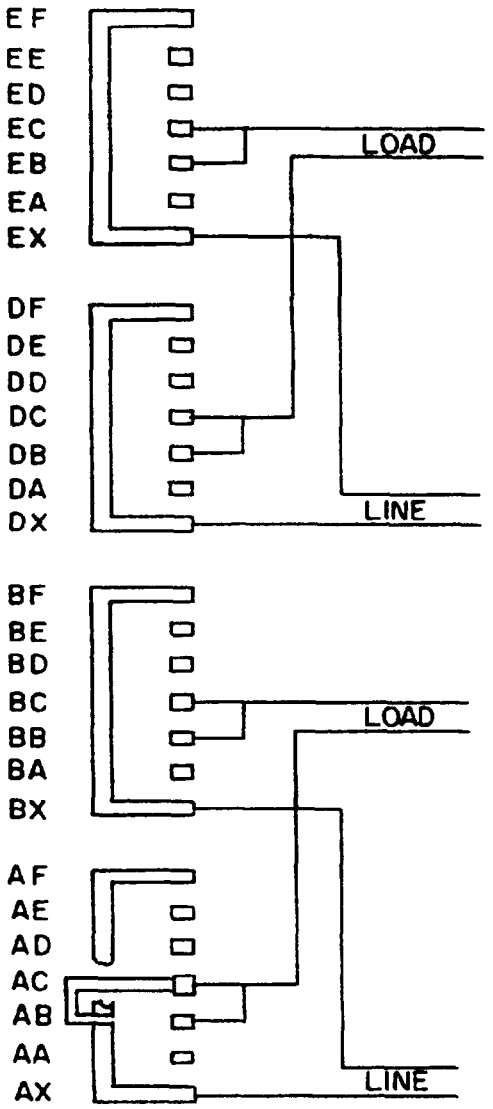
Qualification samples: Two remote operated switch assemblies shall be submitted for qualification inspection.

NOTE: Manufacturers who have qualified to MIL-S-24187/4 are qualified by similarity.

Tests:

Loading of switches: Loading of switches for endurance and temperature rise tests for qualification and quality conformance, group C testing, shall be as shown on figure 7 and table VIII

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NOTE: Sections 1 and 2, row 1, shall be loaded at 2 amperes resistive at 15 volts dc.

Sections 1 and 2, row 15, shall be loaded at 25 microamperes resistive at 1.5 volts dc.

FIGURE 7. *Test loading.*

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Temperature rise: Contacts to be monitored during temperature rise tests are specified in table VIII.

TABLE VIII. *Temperature measurements*

Section	Row	Contacts
1	1	GX, GB, CX, CB
2	1	EX, EB

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy – SH
(Project 5930-N681-10)

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